Presbycusis

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The inevitable deterioration in hearing ability that occurs with age—presbycusis—is a multifactorial process that can vary in severity from mild to substantial. Left untreated, presbycusis of a moderate or greater degree affects communication and can contribute to isolation, depression, and, possibly, dementia. These psychological effects are largely reversible with rehabilitative treatment. Comprehensive rehabilitation is widely available but underused because, in part, of social attitudes that undervalue hearing, in addition to the cost and stigma of hearing aids. Remediation of presbycusis is an important contributor to quality of life in geriatric medicine and can include education about communication effectiveness, hearing aids, assistive listening devices, and cochlear implants for severe hearing loss. Primary care physicians should screen and refer their elderly patients for assessment and remediation. Where hearing aids no longer provide benefit, cochlear implantation is the treatment of choice with excellent results even in octogenarians.

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Introduction

Presbycusis, literally elder hearing, is the general term applied to age-related hearing loss. The term encompasses all conditions that lead to hearing loss in elderly people.[1] The disorder is characterised by reduced hearing sensitivity and speech understanding in noisy environments, slowed central processing of acoustic information, and impaired localisation of sound sources. As a result, people with the disorder have difficulty, proportional to the degree of hearing impairment, in conversation, music appreciation, orientation to alarms, and participation in social activities. There are three classic types of the disorder—sensory, strial, and neural—that can occur alone or in combination. Each type has implications for treatment.

Because of the high prevalence of presbycusis, hearing difficulty is a common social and health problem. Overall, 10% of the population has a hearing loss great enough to impair communication, and this rate increases to 40% in the population older than 65 years.[2] 80% of hearing loss cases occur in elderly people.[3] Although hearing worsens with age, the severity of the hearing problem at any given age varies greatly. It is rare to find a person older than 70 years with no hearing impairment or whose hearing sensitivity has not declined from youthful levels. Hearing levels are poorer in industrialised societies[4] than in isolated or agrarian societies.[5] Thus, it is conceptually useful to regard presbycusis as a mixture of acquired auditory stresses, trauma, and otological diseases superimposed upon an intrinsic, genetically controlled, ageing process.

Presbycusis first reduces the ability to understand speech and, later, the ability to detect, identify, and localise sounds. The loss of hearing sensitivity begins in the highest frequencies, which has an adverse effect on understanding speech in noisy or reverberant places. Once the loss progresses to the 2–4 kHz range, which is important in understanding the voiceless consonants (t, p, k, f, s, and ch), speech understanding in any situation is affected. The most common complaint in presbycusis is not that the patient cannot hear, but rather that they cannot understand what is being said. For example, people will confuse “mash”, “math”, “map”, and “mat”, or “Sunday” with “someday”. Even such minor misperceptions, left uncorrected, can lead to communication errors or worse. The hearing loss extends to the lower frequencies with time resulting in poor speech detection as well as poor speech understanding. High-frequency warning sounds, such as beepers, turn signals, and escaping steam, are not heard or localised, with potentially disastrous results.

Hearing loss affects an individual’s psychosocial situation. Untreated hearing impairment contributes to social isolation, depression, and loss of self-esteem. Hearing impairment has also been implicated as a cofactor in senile dementia,[6] although others have contested this association.[7] Many people regard presbycusis as an inevitable rite of passage into their senior years and are reluctant to seek help because of cost, vanity, and inconvenience. Others may ascribe their problem to mumbling speakers; for these people, recognition and acceptance of an age-related impairment is a difficult psychological hurdle.

Presbycusis management consumes an increasing portion of healthcare expenditures given the rising mean age of people in industrialised societies. Modern hearing aids are valuable aids to communication. Although they cannot restore lost sensory cells, they do provide acoustic power for declining metabolic function. On the horizon are investigations of treatment modalities that might correct the pathophysiological deficits of presbycusis and other hearing disorders. Such modalities will need more accurate diagnosis at the cellular level and greater medical involvement in their use than at present. The primary management of people with presbycusis in most communities is done by those who sell hearing aids. Only about 20% of people who might benefit from amplification have actually purchased an aid[8,9] and 25–40% either underuse or abandon hearing aid use.[10,11] Thus new treatment and management strategies need to be examined.

We review herein the structure and function of the ear, the altered functions in presbycusis, diagnostic considerations, and treatment options. We attempt to

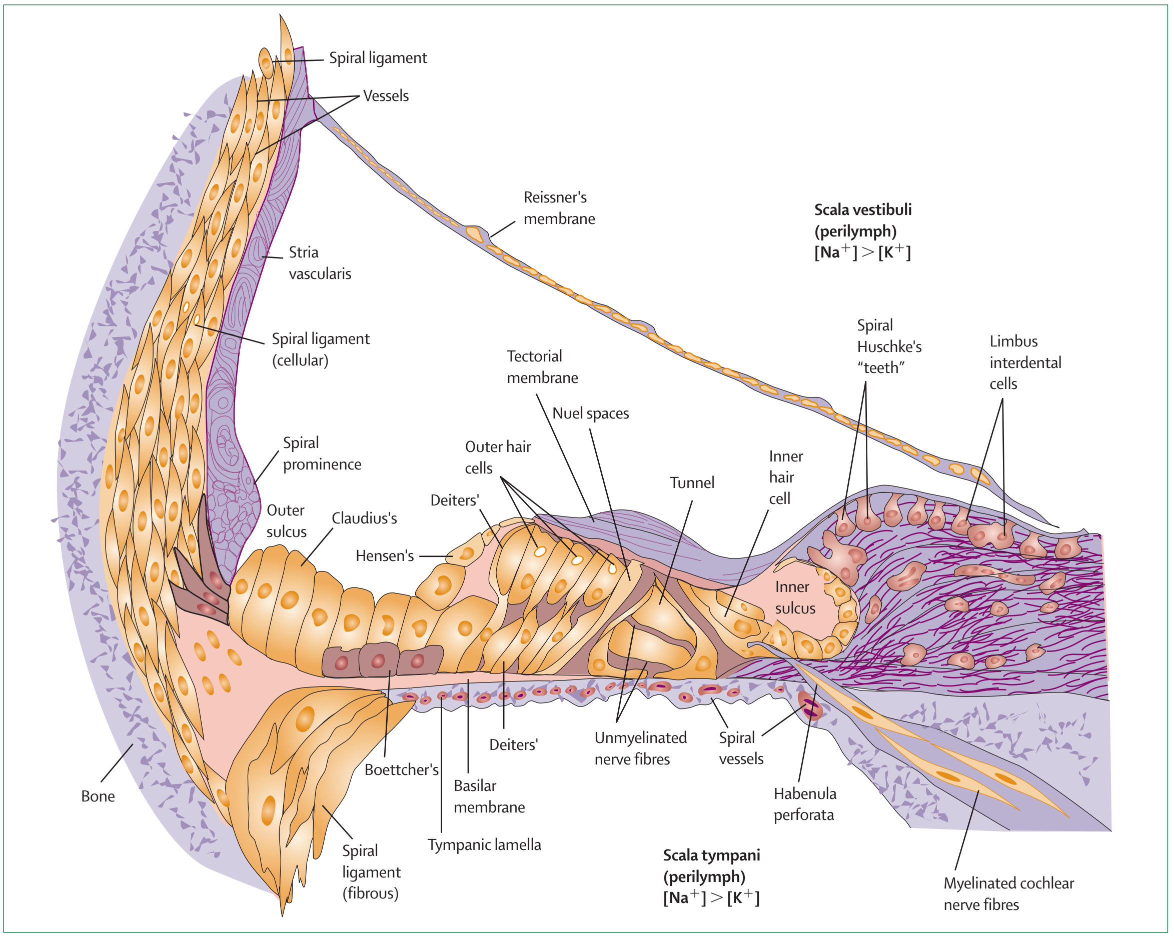

Figure 1: **Cross-section of the cochlea**
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clarify important issues related to the causes of presbycusis and highlight what the practising physician could do to advise patients about the diagnosis, prevention, and treatment of the disorder. Finally, we point out areas of controversy and misunderstanding of the disorder and its treatment, and mention emerging research that could alter the treatment landscape of this highly common problem. In general, knowledge about presbycusis comes from animal models, clinical experience, human temporal bone research, and epidemiological studies. We have drawn from all these sources and have relied upon our extensive personal reference databases for this review.

Definitions

Presbycusis (or presbyacusis) is a general term that refers to hearing loss in the elderly and, as such, represents the contributions of a lifetime of insults to the auditory system. Of these, ageing and noise damage are the chief factors, plus genetic susceptibility, otological disorders, and exposures to ototoxic agents. Some people refer to presbycusis as hearing loss due solely to ageing. Because it is very difficult to isolate age effects from other contributors to age-related hearing loss, we use presbycusis and age-related hearing loss synonymously. There is uncertainty about the mechanisms involved in auditory ageing. Furthermore, there is incomplete and contradictory evidence about the interactions between ageing and noise trauma. Both the peripheral and central auditory pathways are affected in presbycusis and the clinical findings often represent a mixture of abnormalities. Because of space considerations, this review will emphasise peripheral presbycusis, which is the most common problem, and mention central (neural) presbycusis only in passing.

Background

Anatomy and physiology

The external ear comprises the pinna and external auditory canal, which acts as a resonator and enhances sound transmission. The middle ear transforms air vibrations into the fluid-filled inner ear (cochlea) providing a pressure gain of 25–30 dB. The frequency range of human hearing is from 20 Hz to 20 kHz.

Although the structures of the middle ear undergo age-related changes, there is very little effect, if any, on their function,[12] with the notable exception of cerumen production, which seems to increase with age. Whereas the middle ear is a passive device that is essentially unaffected by the ageing process, the cochlea is an active device, with non-linear characteristics, that is dramatically affected by ageing.

The cochlea is a coiled bony tube about 35 mm in length with three compartments: scala tympani, scala vestibuli, and scala media (figure 1).[13] Scalae tympani and vestibuli contain perilymph, which has a high sodium concentration. Scala media contains endolymph with a high concentration of potassium. A direct current resting potential (endolymphatic potential) of 80–90 mV is measurable in scala media. This large direct current resting potential arises from Na^+ K^+ ATPase pumps in the stria vascularis located on the lateral wall of the cochlea. The lateral wall (stria vascularis and spiral ligament) and the direct current resting potential are seriously affected by the ageing process and will be discussed in more detail later.

The auditory transducer is the organ of Corti, which contains sensory cells (three rows of outer hair cells and one row of inner hair cells). Deflection of stereocilia (hairs) of the sensory cells by a mechanical travelling wave initiates the transduction process. A travelling wave along the basilar membrane, moving from the base towards the apex of the cochlea, arises in response to the piston-like movements of the stapes in the middle ear. The travelling wave has a sharply tuned peak that is located basally for high-frequency sounds and progresses apically as the frequency is decreased. Deflection of stereocilia by the travelling wave opens and closes ion channels, resulting in a current flow (K^+) into the sensory cell. The potassium flux arises from the positive 80–90 mV endocochlear potential of scala media added to the negative intracellular potential of outer and inner hair cells. The resulting depolarisation causes an enzyme cascade, releasing chemical transmitters and subsequently activating afferent nerve fibres. Although the notion of the cochlea and the organ of Corti as an active rather than passive organ is no longer debated, specific details of the cochlear amplifier and the biological basis of its operation are under investigation. This research is important because many forms of hearing loss involve the cochlear amplifier, whose non-linear properties allow the inner ear to respond to an extraordinarily wide range of sound intensities while maintaining excellent frequency specificity.

Each auditory nerve contains about 30 000 neurons connecting the sensory cells to the auditory brainstem (cochlear nucleus). The cell bodies of the auditory nerve are located in the central core (modiolus) of the cochlea. Each inner hair cell has ten to 20 dendritic connections. The auditory nerve has two types of fibres: 90–95% are type 1 fibres, which are large, myelinated, bipolar neurons that innervate inner hair cells; the remaining 5–10% are type 2 fibres, which are small, unmyelinated neurons that provide efferent synapses with outer hair cells.

Cochlear ageing in animals

Examination of the inner ears of animals raised in quiet environments provides important documentation of the pathophysiology of presbycusis. These experiments show that degeneration of the stria vascularis is the most prominent element.[14–16] This degeneration usually originates in both the base and apex of the cochlea, extending to mid-cochlear regions as age increases. In some cases, the degeneration is patchy. In addition to age-related systematic degeneration of marginal and intermediate cells of the stria vascularis, there is a loss of Na^+ K^+ ATPase. Sometimes the loss of this important enzyme, which is detectable with immunohistochemical techniques, occurs when the stria vascularis appears normal or nearly normal under light microscopic examination. Accordingly, the prevalence of strial or metabolic presbycusis may prove to be substantially higher in human beings when the appropriate immunohistochemical techniques are applied to the study of temporal bones of older human beings.

The stria vascularis is heavily vascularised and has an extremely high metabolic rate. Histopathological studies of ageing gerbils have provided strong evidence for vascular involvement in age-related hearing loss.[17] Morphometric analyses of lateral wall preparations stained to contrast blood vessels (figure 2) have shown losses of strial capillary area in aged animals. The vascular pathological changes first occurred as small focal lesions mainly in the apical and lower basal turns and progressed with age to encompass large regions at both ends of the cochlea. Remaining strial areas were highly correlated with normal microvasculature and with the endocochlear potential. Not surprisingly, areas of complete capillary loss invariably correlated with regions of strial atrophy. Subsequent ultrastructural analysis has revealed a significant thickening of the basement membrane, which is accompanied by an increase in the deposition of laminin and an abnormal accumulation of immunoglobulin as shown histochemically. Thus, considerable support exists for the major involvement of strial microvasculature in age-related degeneration of stria vascularis. It is important to note that laboratory studies in many animal species and histopathological studies of human temporal bones are consistent in showing age-related degeneration of the stria vascularis.

Age-related degeneration of the stria vascularis has a substantial effect on the basic physiology of the cochlea, especially on the endolymphatic potential, which provides a voltage to the cochlear amplifier. Thus, when the endolymphatic potential is reduced significantly, the operation of the cochlear amplifier is affected. Indeed,

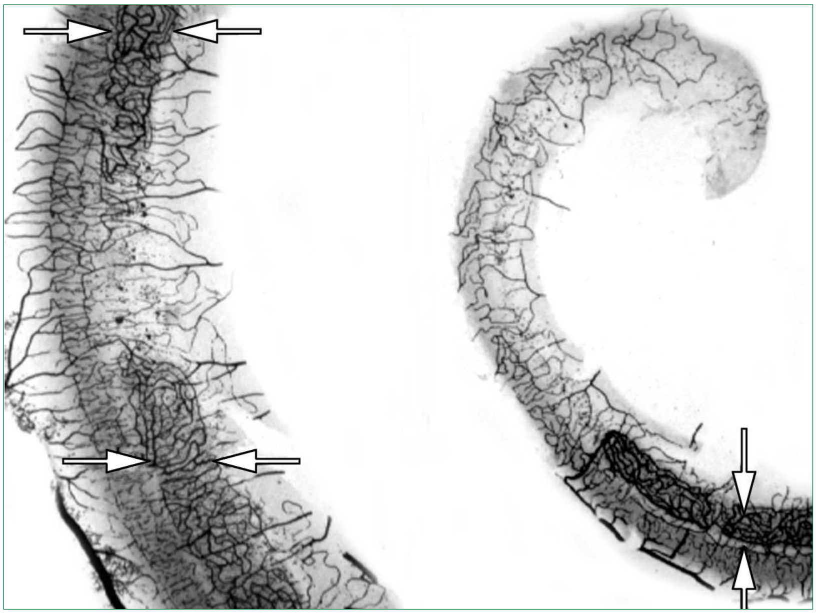

Figure 2: **Surface preparations of lateral wall dissections from an old gerbil stained to contrast blood vessels**
The strial capillary bed (between arrows) overlies vessels of the spiral ligament. A: focal area of strial capillary atrophy. B: complete loss of strial capillaries throughout the apical turn.

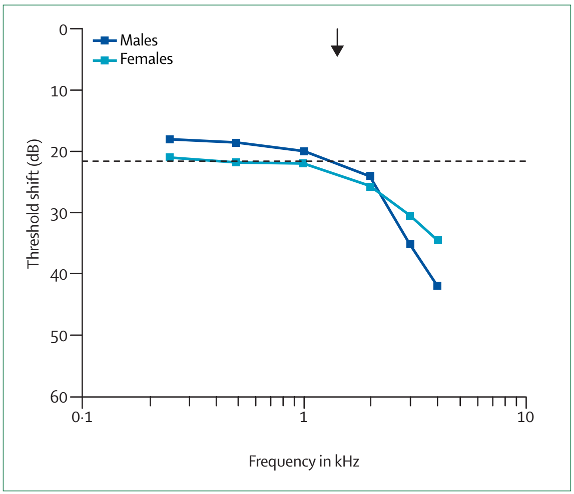

Figure 3: **Threshold shifts in ageing in non-noise-exposed human patients**
Reprinted with permission of The Society for Neuroscience.[18]

when the endolymphatic potential decreases to values of 20 mV or lower the cochlear amplifier is deemed to be voltage starved with a maximum reduction in its gain. The gain ranges from 20 dB in the apex of the cochlea and increases to as much as 60 dB in the base. The audiogram resulting from a voltage-starved cochlear amplifier[18] measured in laboratory animals with endolymphatic potentials less than 20 mV corresponds closely to audiograms recorded in ageing human patients (figure 3)[18] with no history of exposure to noise. As much as 20–30% of the stria could degenerate with only a 20 mV reduction in the endolymphatic potential. As strial degeneration exceeds 50%, endolymphatic potential values drop substantially.

There is a unique feature of strial degeneration: in an ageing animal with a 40–50 dB hearing loss and an endolymphatic potential of 20 mV (normal=90 mV) hearing thresholds can be improved by 20–25 dB by introducing direct current into scala media, which raises the low endolymphatic potential. Direct current, introduced into scala media by the researcher, returns the endolymphatic potential to nearly normal values, reduces the extent of the hearing loss as indicated by the threshold of the compound action potential (CAP), and substantially increases the size of the CAP produced by moderate and high-level signals. Degeneration of the stria vascularis, which has been called the battery of the cochlea, and the resultant decline in the endolymphatic potential, has given rise to the dead battery theory of presbycusis.[17] Indeed, the success of these current-injection experiments and the seeming validity of this theory has produced an engineering effort directed at the development of a direct current hearing aid.[19]

In addition to the age-related decrease in the endolymphatic potential of scala media, there are other changes in the physiology of the ageing cochlea and auditory nerve. One such change is loss of auditory nerve function as indicated by increased thresholds of the CAP of the auditory nerve. Slopes of input-output functions of the CAP in ageing animals are decreased even when the loss of auditory thresholds is only 5–10 dB. In other words, as the signal intensity is increased the amplitude of the CAP increases by a fraction of that recorded in young animals. These shallow input-output functions of the CAP are also reflected in shallow input-output functions of physiological potentials arising from the auditory brainstem. Thus, what seems to be abnormal function of the auditory brainstem in older animals and human beings is only the abnormal output of the auditory nerve. The reduced amplitudes of action potentials recorded in ageing ears probably indicate asynchronous or poorly synchronised neural activity in the auditory nerve. The pathological basis of asynchronous activity in the auditory nerve is unknown, but probably involves the nature of the synapse between individual auditory nerve fibres and the attachment to the inner-hair cell, primary degeneration of spiral ganglion cells, and reductions in the endolymphatic potential. Malfunctions of the auditory nerve caused by a reduced endolymphatic potential remain difficult to differentiate from malfunctions caused by degeneration of spiral ganglion cells.[20] Age-related asynchronous activity of the auditory nerve probably contributes to age-related declines in temporal resolving abilities, which are often attributed solely to age-related declines in the properties of the auditory central nervous system.

Clinical pathophysiology

Loss of threshold sensitivity in the high-frequency region of the hearing spectrum is the first sign of presbycusis. Such changes can begin in young adulthood, but are initially evident at 60 years for the most part. Over time, the threshold elevation progresses to lower and lower frequency areas. This pattern of progression noted in quiet-aged gerbils (figure 4)[21] is typical of declining metabolic function of the cochlea—ie, the stria vascularis[21]—and is termed strial presbycusis.[22] Many people also have loss of outer hair cells in the basal cochlea. Such loss of hair cells, termed sensory presbycusis, is more likely to be the result of specific disorders, usually damage from excessive noise. The audiometric pattern of sensory presbycusis is that of a steeply sloping high-frequency loss, often with a notch or dip in the 4 kHz region.

Other factors, such as genetic susceptibility, age-related diseases, and ototoxic medication, can add to the burden. Hearing loss accumulates from the effects of these various factors, but not necessarily in a linear manner. For example, age changes in hearing accelerate with time whereas the hearing loss from noise tends to decelerate with time. That societal noise contributes to presbycusis is lent support by Goycoolea and colleagues[4] who showed that the mild hearing loss with age of people living their entire lives in the quiet of Easter Island was less than their kinsmen living on the noisier mainland for periods as short as 3–5 years.

Because of difficulties in controlling environmental ototoxicity in human beings and the notable similarities in the structure and function of the ear across mammalian species, animal studies have been vital in understanding the effects of noise and age on the inner ear. Mills and co-workers[21] showed that gerbils raised in quiet have just as much, or more hearing loss with age than groups of noise-reared animals. The variability of the loss was far greater in the quiet-raised group than in the noise-reared group. The interaction of the effects of noise and ageing are not fully understood because, in part, noise and ageing both affect the high frequency regions of the cochlea first. However, noise damage is typified by threshold elevation in the 3–6 kHz range, whereas the earliest effects of age are seen at the highest test frequencies (usually 8 kHz) with a monotonic pattern of loss that is proportionately less at each lower test frequency.

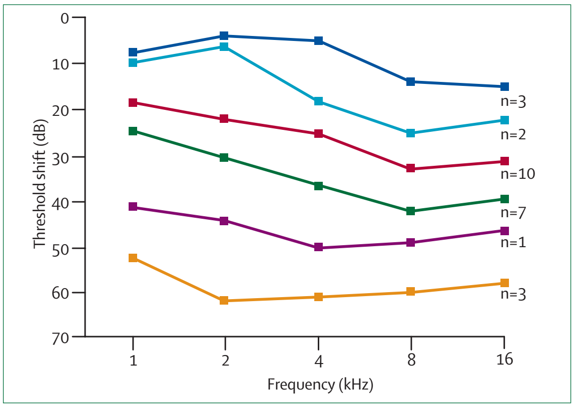

Figure 4: **The progression of age-related threshold shifts in quiet-reared gerbils at frequencies from 1kHz to 16 kHz**
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Human pathology

Early knowledge about the pathology of human presbycusis came from the temporal bone laboratory of Schuknecht[23] who described the light microscopic findings in the human inner ear and compared these to premortem hearing tests. Although the histological technique is crude by today's standards, these examinations provided the opportunity to characterise the hearing in temporal bones with a single pattern of cellular loss. Of necessity, the cellular loss had to be far advanced to be detectable; thus, early lesions due to subtle dysfunction could not be detected at the light microscope level. We regard these archetypes as an important beginning in the study of the pathology of presbycusis.

Schuknecht classified presbycusis into four types: sensory (outer hair-cell loss), neural (ganglion-cell loss), metabolic (strial atrophy), and cochlear conductive (stiffness of the basilar membrane). The cochlear conductive subtype is theoretical and will not be discussed further as we believe such cases would be characterised today with modern histochemistry. Later, Schuknecht added two more categories: mixed and indeterminate, the latter accounting for 25% of cases.[24] Others have noted that most cases exhibit a mixture of pathological changes.[25] The audiometric pattern in Schuknecht's sensory cases is typical of noise-induced hearing loss and the histology findings reveal missing hair cells in the area of the hearing loss. Indeed, most of these cases were men with histories of noise exposure. It is now known that ageing alone does not cause outer hair-cell loss.[21] Thus, we emphasise that in human beings, and in many species of animals (excluding some genetic mouse models), sensory presbycusis probably has little to do with age and much to do with accumulated environmental noise toxicity.

We now know that the pure-tone threshold audiogram is a poor indicator of neural loss since any pattern from normal to anacousis can be seen. The sine qua non of neural lesions is reduced word recognition. Better clinical assessment techniques, such as auditory brainstem response audiometry and otoacoustic emission testing, provide additional detail about neural function. Primary dysfunction of the auditory nerve (auditory neuropathy)[28] is more evident in children than in adults. In this condition, outer hair-cell function, as measured by otoacoustic emissions, is normal but the auditory brainstem responses are reduced or absent. The term auditory dyssynchrony is now used to describe the condition.

We are in agreement with the observations and conclusions of Schuknecht and Gacek[24] that age-related degeneration of the stria vascularis is the most prominent anatomical characteristic of age-related hearing loss. Additionally, the audiometric pattern resulting from strial degeneration in animals is also the most typical finding in cohort studies of elderly people.[10,27–30] By contrast, the steeply sloping audiogram of people with confirmed noise-induced hearing loss[31] differs greatly from the strial pattern and coincides with Schuknecht's sensory presbycusis pattern,[32] which can occur at any age. Indeed, many of his patients had clear histories of noise exposure. In the absence of any clinical method to measure the endocochlear potential in human beings, audiometric patterns are the only way to infer the probable pathophysiology.

In families, the strial pattern of hearing loss had a high heritability index (0·46).[33] Heritability was significant in pairings of women (sisters 0·53, mother– daughters 0·36, siblings 0·38). Sensory presbycusis also had a significant heritability index for both sexes, but at lower correlations than strial presbycusis. Poor low–frequency hearing, which is typical of strial presbycusis, has also been shown to be associated with cardiovascular diseases (heart attacks, stroke, intermittent claudication) and to affect both men and women, but the effect is stronger in women.[34]

Central presbycusis

Given the complex neural connections through which auditory stimuli pass to be perceived in the auditory cortex, it is logical to expect that lesions of these pathways would affect hearing. Indeed, secondary degeneration of central pathways after loss of sensory cells in the cochlea, although slow, is a limiting factor in cochlear implantation in people with long-standing deafness. Primary neural lesions of the auditory pathways are distinctly uncommon, incompletely documented, and poorly understood. Whether primary degeneration and secondary degeneration can co-exist is not known. It is our view that secondary degeneration is the major cause of central presbycusis.

Age changes in the central auditory system are well known to affect speech perception. Such changes are now termed age-related auditory processing disorder. Other synonyms are central auditory dysfunction, neural presbycusis, central presbycusis, and nerve deafness. These changes typically affect speed of processing and result in poorer speech understanding in noise or with rapid or degraded speech. Given that the ultimate perception of speech is in the brain, it is not unexpected that age-related brain dysfunction would affect hearing. Fortunately, central presbycusis severe enough to limit rehabilitation is uncommon and few audiologists include central testing in the assessment of people for amplification candidacy because, in part, of complexity and uncertainty about the implications of the results.

Unfortunately, the term nerve deafness is firmly entrenched in the lay literature and by popular use even though it has led to the frequent misunderstanding that hearing aids will not help. Although people with central presbycusis do indeed have poorer speech understanding than expected from the audiogram, most people with poor speech understanding have outer hair-cell loss rather than neural lesions and are helped by amplification. Fortunately, isolated central presbycusis is uncommon and most cases with reduced speech discrimination also have loss of cochlear sensory cells. We could suspect central dysfunction in about 12% of the cases in the Framingham Heart Study Cohort using as a criterion a significant discordance between measured and predicted speech perception in quiet.[35] Using multiple central auditory tests in the same cohort in an earlier study of 1026 people aged from 64 years to 92 years, we found one abnormal test in 23%, two abnormal tests in 4%, and three abnormal tests in none.[36] None of these people had peripheral hearing loss severe enough to confound the central testing. The prevalence of abnormal test results increased significantly with age but age only accounted for 13% of the variance. We conclude that isolated central presbycusis is not common and that it is difficult to separate the effects of peripheral from central abnormalities.

Diagnosis

History

Hearing loss is often a silent disorder, characterised more by what is missed than what is heard. In many cases, family and friends are more aware of the problem than the patient. Nonetheless, most people with presbycusis will respond to a direct inquiry with a positive answer. People with depression and cognitive dysfunction should be assessed to exclude occult hearing loss as a contributing factor. The hearing loss is often accompanied by tinnitus—the perception of a ringing sound in the ears or head. Tinnitus is a sign of hearing loss and should be assessed to exclude treatable diseases (ie, acoustic neuroma) as a cause.

Risk factors for presbycusis are: 1) noise exposure; 2) smoking; 3) medication; 4) hypertension;[37] and 5) family history. People with substantial exposure to workplace noise, recreational noise, and gun-shooting are more likely to have high-frequency hearing loss. Smoking is also associated with high-frequency loss.[38] The use of aminoglycoside antibiotics, cisplatin, loop diuretics, or anti-inflammatory agents may contribute to hearing loss. Presbycusis does cluster in families.[33]

Physical examination

The physical examination is usually normal after removal of cerumen, which is a common problem in elderly people and a frequent cause of hearing loss and hearing aid malfunction. Topical sodium bicarbonate solution, 10%, is an excellent cerumenolytic. Opacification of the normally translucent tympanic membrane is commonly seen. This has no effect on conduction of sound energy and is simply a manifestation of age.

Screening

Given the high prevalence of presbycusis in people of retirement age and the adverse effects of hearing loss on well-being, screening for hearing loss should be done at annual physical examinations or at the first visit for new patients over the age of 60 years. A single question on an intake form "do you have a hearing problem?" is a very cost effective and sensitive instrument to screen for presbycusis.[39] The 10-item hearing handicap inventory for the elderly-short form (HHIE-S) is widely used for screening. However, the HHIE-S under-reports hearing loss because its sensitivity is lower than the single question "do you have a hearing problem".[39] The value of screening is justified by the effectiveness of remediation.[40]

The use of clinical measures of hearing loss, such as spoken voice tests and finger friction tests, are imprecise and are not effective or reliable methods for screening. Screening audiometry administered by a trained office nurse or medical assistant is a practical and cost-effective method for detecting significant hearing loss.[41] The equipment needed for screening audiometry is light weight, low cost, and well accepted by patients.[42] The standard screening audiometer tests at 1 kHz, 2 kHz, and 3 kHz at intensity levels of 25 dB, 40 dB, and 60 dB. Failure at any one frequency at 25 dB for younger adults or 40 dB for retired individuals justifies a referral for definitive assessment.

Imaging is not done except where the loss of hearing is unilateral or significantly asymmetric, or where tinnitus is unexplained by the audiogram. A metabolic assessment might be indicated if the patient has not had a recent health examination. Diabetes, renal dysfunction, hypertension, and hyperlipidaemia should be excluded as cofactors.

Central auditory testing

A variety of central auditory tests are available but these are rarely done in the clinical setting unless there is a large discrepancy between the history and the results of standard peripheral auditory tests. For example, when a patient has reasonable speech understanding in quiet but has severe speech problems in noisy or difficult listening environments, central presbycusis is likely to be a factor. Unfortunately, contemporary rehabilitation measures—other than environmental manipulation—are ineffective in dealing with central problems. Moreover, few tests have any discrete localising value and are not done as part of the neurological workup. The most widely used tests assess speech recognition in relation to the pure-tone thresholds and to noise.

The simplest test is auditory rollover.[43] Speech understanding normally improves as the level of the presentation of the phonetically balanced words increases until distortion due to overstimulation of the cochlea occurs. People with central presbycusis may exhibit a 20% or greater decrease in speech recognition at high signal intensities. Rollover was very uncommon (1% of patients) in our survey of the older members of the Framingham Heart Study Cohort.[36] Some people with loudness recruitment, a common problem with cochlear hearing loss, have difficulty in performing the rollover test because at high presentation levels the apparent loudness of the sound increases substantially.

A widely available central auditory test is the synthetic sentence identification with either an ipsilateral competing message (SSI-ICM) or a contralateral competing message (SSI-CCM).[44] The SSI-ICM seems to be more sensitive.[45] In this test, the patient listens to a narrative about Davy Crockett and is instructed to listen for one of ten sentences superimposed on the narrative by the same speaker. Each sentence has no meaning even though each series of three words makes syntactical sense. The test is quite easy for most people even when the signal and the message are presented at the same intensity level, usually 40 dB above threshold. Since audibility is not an issue, the test requires the listener to attend to the message and ignore the narrative.

We have shown that poor performance on the SSI-ICM is common in people with probable Alzheimer's disease[46] and that very poor performance (<50% correct) in either ear might precede the clinical onset of dementia of the Alzheimer's type by several years.[47] We presume that frontal lobe executive control function is a key factor in the task because lesions of the central auditory pathway in Alzheimer's disease are uncommon in early cases.[48] With the possibility of treatments to delay the progression of Alzheimer's disease, early identification assumes great importance. Referral to an otology centre can assess this possibility.

Many other tests are available. Two more widely available tests are the speech perception in noise (SPIN) test[49] and the staggered spondaic words test (SSW).[50] Both use carefully prepared speech materials to assess central auditory function.

Treatment

Hearing loss of all types affects not only communication but also quality of life. Mulrow and co-workers[51] studied the effect of amplification on older patients with hearing loss and documented a positive effect of personal amplification (hearing aids) on quality of life.[51] Therefore, treatment effects extend beyond communication. No treatment exists now to restore lost hearing. Research into hearing restoration is a growing scientific field.

Prevention

Although some degree of sensory presbycusis is inevitable, the deterioration can be reduced by avoidance of hazardous noise exposure or use of suitable hearing protection. Insert ear plugs provide about 15–25 dB of attenuation and can thus permit people to work in otherwise hazardous areas. The effects of noise accumulate over a lifetime; gun shooting and loud music exposure during adolescence will contribute to communication difficulties during the retirement years.

Cardiovascular disease and its risk factors affect hearing to some extent. Stroke, myocardial infarction, claudication, hypertension, hyperlipidaemia, and diabetes mellitus have all been associated with excessive hearing loss. Therefore, that maintenance of good general health and fitness would reduce the risk of hearing loss due to systemic disease would seem logical. High lipid diets are associated with poor hearing.[52] There is inconclusive evidence that low-caloric diets, which clearly prolong life in laboratory animals, have any effect on presbycusis.[53] While free radical accumulation is implicated in presbycusis, anti-oxidant agents have not been shown to counter auditory ageing.

Rehabilitation

Communication courtesy

Communication is a two-way process; the burden of communication falls equally on the speaker and listener. Common courtesy holds that both should work at improving the communication environment as well as the process when one of them has a hearing problem. The speaker should: be face-to-face with the listener, speak clearly and unhurriedly, turn off competing sound sources (TV, radio), and make sure that the message was received. The hearing-impaired listener should also be serious about communicating and take steps to repeat what was heard so misunderstandings can be corrected. These principles apply to all degrees of hearing losses.

Environmental manipulation

Central nervous system processing of speech slows down with age. Older people have more difficulty in understanding speech than do younger people, even when cochlear sensitivity is similar. Degraded speech signals, such as those that occur in noise, in reverberant halls, or with rapid speakers, are more difficult to understand. Such problems are thought to indicate slowing of the central auditory system's ability to integrate and synthesise speech sounds into meaningful language elements. Therefore, optimising the listening environment, often by simple means (turning off the radio, speaking slower), has important effects on speech comprehension.

Amplification

As a rule of thumb, when the average hearing thresholds reach 40 dB on the audiogram, amplification is indicated. People with smaller losses also receive benefit, especially when employment or educational needs are considered. The choice of types of hearing aids and models is staggering. Hearing aids may have analogue or digital circuitry, may be adjusted manually or automatically, and may have sophisticated directional microphones, noise suppression technology, telephone coils, and multiple programme modes (for quiet, noise, music). The smaller the physical size of the device, the greater the cost.

Although amplification technology has made great strides in recent years, controversy remains regarding the benefits of these various choices, especially in relation to cost.[54,55] Yeuh and colleagues[56] have shown a clear advantage for digital technology in hearing aids. Notable advances in directional microphones and noise-suppression circuitry in high-end hearing aids have greatly extended their benefit.[57]

Hearing aids have many drawbacks. They do not restore normal hearing. They need a long learning and adjustment period in which the brain adapts to the new way things sound. They are uncomfortable, unsightly, and costly. Nonetheless, until such time in the distant future that a biological correction for presbycusis is developed, they are the mainstay of rehabilitation for moderate or worse hearing loss. The greater the loss, the more the individual relies on amplification. People with mild high-frequency losses are often annoyed with the amplification of sounds such as paper shuffling and their own breathing, and, as a consequence, use the aids intermittently. Such a usage pattern prevents central adaptation to amplified sounds and contributes to user dissatisfaction. Reliance on education and environmental manipulation is more likely to succeed in the early case. Education about reasonable expectations enhances hearing aid acceptance.

Speech reading

Aural communication is enhanced by viewing the speaker's face. Given that speech is often redundant, facial expressions and lip contours provide assistance in filling the gaps resulting from unheard speech sounds. Formal speech reading classes are available at a few centres. Unfortunately, not enough centres are available to provide adequate speech reading training. Materials are now becoming available on the internet (search engine phrase "speech reading").

Auditory training

For people with severe losses, auditory training is of benefit. The hearing impaired listener is trained to identify speech sounds and key words with amplification in place. Such training is tedious and worthwhile, but is seldom available outside of University centres.

Assistive listening devices

A wide range of assistance is available. Ranging from frequency-modulation transmitters, amplified telephones, teleconnectors to couple hearing aids with the telephone, captioning of televised or live programmes, infrared systems for home television, flashing alarms for the doorbell, telephone, and smoke detectors. Assistive listening devices are important and increasingly available adjuncts for the severely hearing impaired person.

Cochlear implants

Cochlear implants are indicated at any age for people with bilateral severe hearing losses not materially helped by hearing aids. Current criteria include hearing no better than identifying 50% or fewer key words in test sentences in the best aided condition in the worst ear and 60% in the better ear. People of retirement age generally are excellent implant candidates since language skills are good and duration of deafness is short. Surgical morbidity is low and acceptance is high.

Conclusions

Presbycusis is common. The disorder deprives older people of key sensory input, which seriously affects their quality of life. Modern rehabilitation strategies are effective but underused. Primary physicians, especially those who care for elderly people, should consider the effect of presbycusis on health. Improvement in general health occurs after auditory rehabilitation.[51] Recent evidence suggests that hearing loss may be an early sign of, as well as a contributor to, dementia.[47] Screening for hearing loss in elderly people has health as well as hearing benefits. Modern amplification methods provide improved communication ability for most users. Regeneration biology is a vigorous new research field that is pursuing methods to restore hearing by regrowth of new hair cells[58] and enhancement of strial dysfunction by chemical and prosthetic means.[19] These promising efforts for future treatments await further research and development.

Conflict of interest statement

G A Gates is a consultant to Advanced Cochlear Systems.

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